

Direct sums of GNS representations and shell unitaries

[Back to Project mathematical routes](#)

Scope

Inequivalent pure-GNS classes form the atomic direct sum. Generic shell theorems and the CAR realization supply faithful irreducible operator models, with the required residual-line hypotheses discharged in their appropriate specialization.

7 direct Cards + 9 reused prerequisites = 16 unique Cards. This count is a selected Card closure, not a source-declaration count.

[Route reading PDF](#) · [Preserved source exploration](#)

Cards in this route

- Key result: Realizing a shell family by a faithful irreducible operator algebra
- Key result: An irreducible operator algebra constructed from projection shells

[Read this route with prerequisites](#)

Detailed route conditions

Let I denote pure-GNS equivalence classes and (H_i, π_i, ξ_i) the root-preserving selected data. Distinct classes, not merely distinct state functionals, give inequivalent representations. Their direct sum is π on $\mathcal{H} = \bigoplus_i H_i$. Write $e_i : H_i \rightarrow \mathcal{H}$ and $b_i = e_i \xi_i$.

The generic atomic-shell theorem takes the rank-one limiting defects as hypotheses. The pure-GNS specialization discharges irreducibility, inequivalence and normalization, while retaining the shell hypotheses. The separate CAR realization proves the required residual-line identities. It chooses unitary links once. The generic generated algebra permits arbitrary bounded extra operators and does not itself assert the CAR consequences.

Dependency-first reading route

Read selected Card prerequisites before their uses. Levels are recomputed from the selected reachability-preserving projection; omitted source helpers remain traceable in the source exploration.

Level 0 (4 Cards)

Level 0

The selected GNS representation of a pure-state class

A choice of representative pure states, fixed literally at a root, gives one concrete GNS representation per equivalence class.

`MathlibAnnex.CStarAlgebra.PureState.selectedRepresentation`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: The matching GNS fiber retains exactly its cyclic line · Distinct GNS classes have no unitary intertwiner · The atomic direct sum of the selected CAR representations

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Level 0

The closed operator algebra generated by a representation and extra operators

Places a represented algebra and a chosen family of bounded operators in one concrete closed algebra.

`MathlibAnnex.CStarAlgebra.AtomicConstruction.concreteTarget`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: An irreducible operator algebra constructed from projection shells

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Level 0

The CAR algebra as a completion of finite matrix stages

Names the completed CAR algebra in which finite matrix calculations can be extended by norm approximation.

`MathlibAnnex.CStarAlgebra.CAR.Limit`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: The product-vector state on the completed CAR algebra · The decreasing root projections in the CAR completion

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Level 0

A fixed representative family of CAR shell data

One family records a transporting automorphism and exact shell links for each selected pure-state class, with an explicitly fixed root component.

`MathlibAnnex.CStarAlgebra.CAR.RepresentativeShellFamily`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: The matching GNS fiber retains exactly its cyclic line · An inequivalent GNS fiber has no residual common range

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Level 1 (4 Cards)

Level 1 An irreducible operator algebra constructed from projection shells Shell partial isometries with rank-one limiting defects can be completed to unitary links that join inequivalent irreducible fibers. MathlibAnnex.CStarAlgebra.AtomicConstruction.exists_irreducible_atomicShellModel
Immediate Card prerequisites: The closed operator algebra generated by a representation and extra operators Used by in this scope: The atomic-shell construction on selected pure-GNS fibers
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Level 1 Distinct GNS classes have no unitary intertwiner The class index makes the selected pure-GNS family pairwise unitarily inequivalent. MathlibAnnex.CStarAlgebra.PureState.no_unitaryIntertwiner_selectedRepresentation
Immediate Card prerequisites: The selected GNS representation of a pure-state class Used by in this scope: An inequivalent GNS fiber has no residual common range · The atomic-shell construction on selected pure-GNS fibers
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Level 1 The product-vector state on the completed CAR algebra Compatible evaluation at the distinguished matrix coordinate extends continuously to the CAR completion. MathlibAnnex.CStarAlgebra.CAR.rootState
Immediate Card prerequisites: The CAR algebra as a completion of finite matrix stages Used by in this scope: Root compression becomes scalar in norm · Purity of the completed root state
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Level 1 The decreasing root projections in the CAR completion The distinguished stage corners give a nested sequence of projections supporting the root state. MathlibAnnex.CStarAlgebra.CAR.rootFlag
Immediate Card prerequisites: The CAR algebra as a completion of finite matrix stages Used by in this scope: Root compression becomes scalar in norm
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Level 2 (3 Cards)

Level 2 The atomic-shell construction on selected pure-GNS fibers Pure-state GNS data supplies the irreducible and inequivalent fibers required by the generic shell construction. MathlibAnnex.CStarAlgebra.AtomicConstruction.exists_irreducible_pureAtomicShellModel Immediate Card prerequisites: Distinct GNS classes have no unitary intertwiner · An irreducible operator algebra constructed from projection shells Used by in this scope: Realizing a shell family by a faithful irreducible operator algebra Open Declaration Card Read exact source
Level 2 Root compression becomes scalar in norm Extends an exact rank-one compression identity on finite stages to a pointwise norm limit on the completed algebra. MathlibAnnex.CStarAlgebra.CAR.tendsto_norm_compressionError Immediate Card prerequisites: The product-vector state on the completed CAR algebra · The decreasing root projections in the CAR completion Used by in this scope: The matching GNS fiber retains exactly its cyclic line · An inequivalent GNS fiber has no residual common range Open Declaration Card Read exact source
Level 2 Purity of the completed root state Passes extremality of the root vector states from every finite matrix stage to the completed CAR algebra. MathlibAnnex.CStarAlgebra.CAR.isPureState_rootState Immediate Card prerequisites: The product-vector state on the completed CAR algebra Used by in this scope: The matching GNS fiber retains exactly its cyclic line · The atomic direct sum of the selected CAR representations · An inequivalent GNS fiber has no residual common range Open Declaration Card Read exact source

Level 3 (3 Cards)

Level 3 The atomic direct sum of the selected CAR representations All selected pure-GNS fibers of the completed CAR algebra act together on one Hilbert direct sum. MathlibAnnex.CStarAlgebra.CAR.selectedAtomicRepresentation
Immediate Card prerequisites: The selected GNS representation of a pure-state class · Purity of the completed root state Used by in this scope: Realizing a shell family by a faithful irreducible operator algebra
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Level 3 The matching GNS fiber retains exactly its cyclic line Identifies the residual projection of a transported CAR flag in its matching pure-state representation. MathlibAnnex.CStarAlgebra.CAR.selected_commonFixedProjection_eq_rankOne
Immediate Card prerequisites: Root compression becomes scalar in norm · The selected GNS representation of a pure-state class · Purity of the completed root state · A fixed representative family of CAR shell data Used by in this scope: The atomic common range is one embedded GNS line
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Level 3 An inequivalent GNS fiber has no residual common range Uses compression and cyclic transport to exclude fixed vectors in every other chosen pure-state class. MathlibAnnex.CStarAlgebra.CAR.selected_commonFixedProjection_eq_zero
Immediate Card prerequisites: Root compression becomes scalar in norm · Distinct GNS classes have no unitary intertwiner · Purity of the completed root state · A fixed representative family of CAR shell data Used by in this scope: The atomic common range is one embedded GNS line
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Level 4 (1 Card)

Level 4 The atomic common range is one embedded GNS line Assembles the matching and inequivalent fiber calculations in an arbitrary Hilbert direct sum. MathlibAnnex.CStarAlgebra.CAR.iInf_range_atomic_transportFlag_eq_span
Immediate Card prerequisites: The matching GNS fiber retains exactly its cyclic line · An inequivalent GNS fiber has no residual common range Used by in this scope: Realizing a shell family by a faithful irreducible operator algebra
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Level 5 (1 Card)

Level 5

Realizing a shell family by a faithful irreducible operator algebra

Constructs unitary links between pure-state summands and an irreducible algebra containing a faithful copy of CAR.

`MathlibAnnex.CStarAlgebra.CAR.exists_completedAtomicShellModel`

Immediate Card prerequisites: The atomic common range is one embedded GNS line · The atomic direct sum of the selected CAR representations · The atomic-shell construction on selected pure-GNS fibers

Used by in this scope: None in this selected scope

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